Zomato Analysis

Capstone Data Analytics Project

-By Mansi Gahlot

About the Zomato

Zomato is one of India's leading food technology companies that has significantly transformed the way people discover restaurants and order food. Founded in 2008 by Deepinder Goyal and Pankaj Chaddah, the company initially started as a restaurant discovery platform named Foodie Bay before rebranding to Zomato in 2010. Headquartered in Gurugram, India, Zomato began by providing restaurant menus and reviews online, helping customers make informed dining decisions. Over time, it expanded its services to include online food delivery, making it a complete digital platform connecting customers, restaurants, and delivery partners.

The core strength of Zomato lies in its technology-driven business model. Through its mobile application and website, customers can browse restaurants based on cuisine, ratings, price range, and location. The platform provides detailed menus, customer reviews, photos, and real-time order tracking, enhancing transparency and user experience. Zomato earns revenue primarily through commissions charged to restaurants on each order, delivery fees from customers, subscription services such as Zomato Gold (now Zomato Pro), and advertising services offered to restaurant partners. Additionally, the company launched Hyper pure, a B2B supply chain initiative that supplies fresh ingredients and kitchen essentials directly to restaurants, further strengthening its ecosystem.

Zomato makes extensive use of data analytics and artificial intelligence to improve its services. It analyses customer preferences, order history, and behavioural patterns to provide personalized recommendations. Advanced algorithms are used to optimize delivery routes, reduce delivery time, and manage demand during peak hours. This technology integration has enabled Zomato to scale rapidly across multiple cities while maintaining operational efficiency.

In 2021, Zomato became one of the first Indian food-tech startups to go public by listing on the NSE and BSE, marking a significant milestone in its growth journey. The company further expanded into quick commerce by acquiring Blinkit, allowing it to enter the instant grocery delivery market. Despite facing strong competition from companies like Swiggy and dealing with challenges related to profitability and delivery management, Zomato has maintained a strong market presence due to its brand value, wide restaurant network, and continuous innovation.

Overall, Zomato has played a crucial role in digitizing the food and restaurant industry in India. By combining technology, logistics, and customer-centric strategies, it has created a comprehensive platform that benefits consumers, restaurant owners, and delivery partners alike. Its evolution from a simple restaurant listing website to a diversified food-tech and quick commerce company highlights its adaptability and growth-oriented vision.

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Aim

The main aim of analyzing the **Indian-Resturants.csv** dataset is to understand the restaurant industry in India by examining important factors such as pricing, ratings, services, and customer convenience features. This dataset helps in studying how different restaurants operate and what facilities they provide to customers.

The analysis focuses on identifying patterns in restaurant types, price ranges, average cost for two, and customer ratings. It also aims to evaluate the availability of services like online delivery, table booking, Wi-Fi, and alcohol. By analyzing these features, we can understand customer preferences, service trends, and which types of restaurants perform better in terms of ratings and popularity.

Another important objective is to explore the relationship between restaurant features and customer satisfaction. For example, we can check whether restaurants offering online delivery or table booking receive higher ratings. This helps in drawing meaningful business insights.

Overall, the aim of this project is to perform data analysis to gain insights into the Indian restaurant market, identify key trends, and support better decision-making for restaurant owners, investors, and business analysts.

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Objective

The primary objective of this project is to perform a detailed analysis of Zomato restaurant data to understand the factors that influence customer preferences and restaurant performance. The study aims to evaluate key variables such as restaurant ratings, customer reviews, pricing (cost for two), cuisine types, and service availability, including online delivery and table booking options. By examining these factors, the project seeks to identify patterns and relationships that impact customer decision-making and overall business success.

Another important objective is to analyze the distribution of restaurants across different locations and determine high-demand and high-competition areas. The project also aims to identify the most popular cuisines and top-performing restaurants based on ratings and customer engagement. Additionally, it focuses on understanding how pricing strategies affect ratings and customer satisfaction.

Through data cleaning, processing, and visualization techniques, the project intends to present insights in a clear and structured format using dashboards and reports. Ultimately, the objective is to provide meaningful, data-driven insights that can support better strategic planning, improve service quality, and enhance customer experience in the restaurant industry.

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Tools

Python

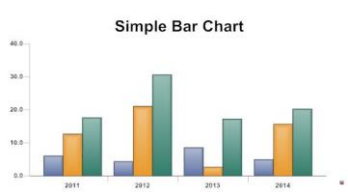
Use:

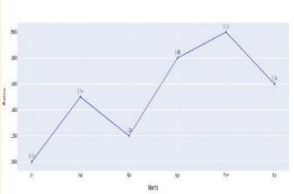
A high-level programming language widely used for data analysis, automation, and AI. Features: Easy to learn and read. Huge library support for data, web, and AI.

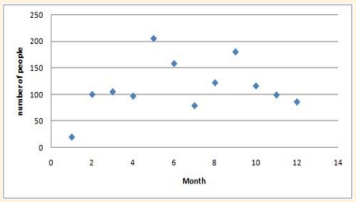
Plotly

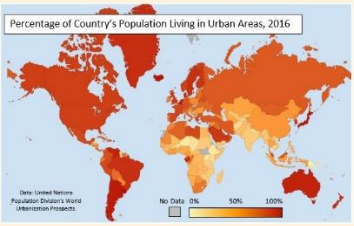
Plotly is an interactive data visualization library used to create dynamic and engaging charts. It supports various visualizations such as bar charts, line charts, scatter plots, pie charts, and maps. Plotly allows users to interact with data through features like zooming, hovering, and filtering.

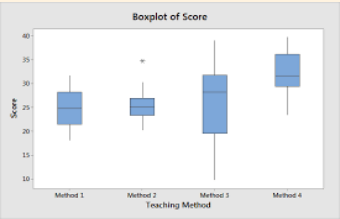
Graphs











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Problem Statement

Task 1 – Explore the basic characteristics of the dataset, including dimensions, data types, and missing values.

Task 2 – Calculate and visualize the average rating of restaurants.

Task 3 – Analyse the distribution of restaurant ratings to understand the overall rating landscape.

Task 4 – Identify the city with the highest concentration of restaurants.

Task 5 – Visualize the distribution of restaurant ratings across different cities.

Task 6 – Determine the most popular cuisines among the listed restaurants.

Task 7 – Investigate if there's a correlation between the variety of cuisines offered and restaurant ratings.

Task 8 –Analyze the relationship between price range and restaurant ratings.

Task 9 –Visualize the average cost for two people in different price categories.

Task 10 – Investigate the impact of online order availability on restaurant ratings.

Taks 11 – Analyze the distribution of restaurants that offer table booking.

Task 12 - Identify and visualize the top restaurant chains based on the number of outlets.

Task 13 – Explore the ratings of these top chains.

Task 14 – Analyze the distribution of restaurants based on features like Wi-Fi, Alcohol availability, etc.

Task 15 - Investigate if the presence of certain features correlates with higher ratings.

Task 16 – Create a word cloud based on customer reviews to identify common positive and negative sentiments.

Task 17 – Analyze frequently mentioned words and sentiments.

Task 18 – Explore if there are any seasonal trends in restaurant ratings or user reviews.

Task 19 – Visualize the distribution of ratings during different times of the year.

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Data Overview

Basic Characteristics of the dataset, including dimensions, data types, and missing values.

```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv("/content/Indian-Resturants.csv")
```

```
df
```

	res_id	name	establishment	url	address	city	city_id	locality	latitude	longitude	...	price_range	currency	highlights	aggregat
0	3400299	Bikanervala	['Quick Bites']	https://www.zomato.com/agra/bikanervala-khanda...	Kalyani Point, Near Tulsi Cinema, Bypass Road,...	Agra	34	Khandari	27.211450	78.002381	...	2.0	Rs.	['Lunch', 'Takeaway Available', 'Credit Card',...	
1	3400005	Mama Chicken Mama Franky House	['Quick Bites']	https://www.zomato.com/agra/mama-chicken-mama-...	Main Market, Sadar Bazaar, Agra Cantt, Agra	Agra	34	Agra Cantt	27.160569	78.011583	...	2.0	Rs.	['Delivery', 'No Alcohol Available', 'Dinner',...	
2	3401013	Bhagat Halwai	['Quick Bites']	https://www.zomato.com/agra/bhagat-halwai-2-sh...	62/1, Near Easy Day, West Shivaji Nagar, Goalp...	Agra	34	Shahganj	27.182938	77.979684	...	1.0	Rs.	['No Alcohol Available', 'Dinner', 'Takeaway A...	
3	3400290	Bhagat Halwai	['Quick Bites']	https://www.zomato.com/agra/bhagat-halwai-civi...	Near Anjana Cinema, Nehru Nagar, Civil Lines, ...	Agra	34	Civil Lines	27.205668	78.004799	...	1.0	Rs.	['Takeaway Available', 'Credit Card', 'Lunch',...	

The dataset was imported using Pandas with the read_csv() function to load the Indian Restaurants data into a DataFrame. Initial exploration was performed to understand the structure, columns, and key variables such as restaurant name, city, price range, and services offered. This step helped in identifying important features required for further data cleaning and analysis.

```
df.dtypes
```

res_id	int64
name	object
establishment	object
url	object
address	object
city	object
city_id	float64
locality	object
latitude	float64
longitude	float64
zipcode	object
country_id	float64
locality_verbose	object
cuisines	object
timings	object
average_cost_for_two	float64
price_range	float64
currency	object

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```
df.shape
```

```
(172638, 26)
```

```
df.describe()
```

	res_id	city_id	latitude	longitude	country_id	average_cost_for_two	price_range	aggregate_rating	votes	photo_count	opentable_support	delivery	take
count	1.726380e+05	172637.000000	172637.000000	172637.000000	172637.0	172637.000000	172637.000000	172637.000000	172637.000000	172637.000000	172629.0	172637.000000	172
mean	1.296643e+07	4190.971536	21.873982	77.622804	1.0	626.316914	1.915974	3.420881	433.330561	306.806519	0.0	-0.245295	
std	8.142839e+06	5454.366854	24.982883	7.190592	0.0	642.702434	0.914563	1.293266	1011.185321	953.563794	0.0	0.967292	
min	5.000000e+01	1.000000	0.000000	0.000000	1.0	0.000000	1.000000	0.000000	-18.000000	0.000000	0.0	-1.000000	
25%	3.000063e+06	9.000000	16.304366	75.327376	1.0	250.000000	1.000000	3.300000	18.000000	4.000000	0.0	-1.000000	
50%	1.869139e+07	29.000000	22.780771	77.301499	1.0	400.000000	2.000000	3.800000	123.000000	24.000000	0.0	-1.000000	
75%	1.880850e+07	11304.000000	26.895069	80.194568	1.0	750.000000	2.000000	4.100000	431.000000	175.000000	0.0	1.000000	
max	1.915979e+07	11354.000000	10000.000000	91.832769	1.0	30000.000000	4.000000	4.900000	42539.000000	17702.000000	0.0	1.000000	

The dataset structure was examined using df.dtypes, df.shape, and df.describe() to understand data types, dimensions, and statistical summary. The dataset contains 1.7+ lakh records with 26 features, including numerical and categorical variables. Descriptive statistics helped identify key insights such as average cost, rating distribution, and vote patterns, forming the basis for further analysis and data cleaning.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 172638 entries, 0 to 172637
Data columns (total 26 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   res_id                172638 non-null  int64
 1   name                  172638 non-null  object
 2   establishment         172638 non-null  object
 3   url                   172638 non-null  object
 4   address               172508 non-null  object
 5   city                  172637 non-null  object
 6   city_id               172637 non-null  float64
 7   locality              172637 non-null  object
 8   latitude              172637 non-null  float64
 9   longitude             172637 non-null  float64
 10  zipcode               43979 non-null   object
 11  country_id            172637 non-null  float64
 12  locality_verbose      172637 non-null  object
 13  cuisines              171350 non-null  object
 14  timings               169961 non-null  object
 15  average_cost_for_two  172637 non-null  float64
 16  price_range           172637 non-null  float64
 17  currency              172637 non-null  object
 18  highlights            172637 non-null  object
 19  aggregate_rating      172637 non-null  float64
 20  rating_text           172637 non-null  object
 21  votes                 172637 non-null  float64
 22  photo_count           172637 non-null  float64
 23  opentable_support     172629 non-null  float64
 24  delivery              172637 non-null  float64
 25  takeaway              172637 non-null  float64
dtypes: float64(12), int64(1), object(13)
memory usage: 34.2+ MB
```

The df.info() function was used to obtain a detailed overview of the dataset, including column names, data types, and non-null values. The dataset contains 172,638 entries and 26 columns with a mix of integer, float, and object data types. It also helped identify missing values in certain columns such as zipcode and timings. This step was essential for understanding data completeness and planning data cleaning procedures.

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Handling the missing values

Sorting missing values means identifying and arranging columns based on the number of null (NaN) values they contain. This helps in understanding which features have the highest data gaps and need immediate attention.

```
# Finding missing values
df.isnull().sum()
```

	0
res_id	0
name	0
establishment	0
url	0
address	134
city	0
city_id	0
locality	0
latitude	0
longitude	0
zipcode	163187
country_id	0
locality_verbose	0
cuisines	1391
timings	3874
average_cost_for_two	0
price_range	0

```
df["address"].fillna("Unknown", inplace = True)
```

```
/tmp/ipython-input-2238304903.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the

  df["address"].fillna("Unknown", inplace = True)
```

```
df["zipcode"].fillna("Unknown", inplace = True)
```

```
/tmp/ipython-input-1541672319.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the

  df["zipcode"].fillna("Unknown", inplace = True)
```

```
df.dropna(subset=['latitude', 'longitude'], inplace=True)
display(df[['latitude', 'longitude']].isnull().sum())
```

	0
latitude	0
longitude	0

dtype: int64

```
df["cuisines"].fillna("Unknown", inplace = True)
df["timings"].fillna("unknown", inplace = True)
```

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```
df["opentable_support"].mean()
```

```
np.float64(0.0)
```

```
df["opentable_support"].fillna("0", inplace = True)
```

```
/tmp/ipython-input-2856965403.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the


  df["opentable_support"].fillna("0", inplace = True)
/tmp/ipython-input-2856965403.py:1: FutureWarning: Setting an item of incompatible dtype is deprecated and will raise an error in a future version of pandas. Value '0' has dtype incompatible with
  df["opentable_support"].fillna("0", inplace = True)
```

After handling all the missing values, the dataset is now completely clean with **zero null values across all columns**. The output of df.isnull().sum() confirms that fields such as address, zipcode, cuisines, timings, latitude, and longitude no longer contain any missing data. This indicates that appropriate data cleaning techniques such as filling null values, replacing them with relevant placeholders, or removing incomplete records have been successfully applied. As a result, the dataset is now consistent, reliable, and ready for further exploratory data analysis, visualization, and model building without the risk of errors or biased insights caused by missing information.

```
df.isnull().sum()
```

	0
res_id	0
name	0
establishment	0
url	0
address	0
city	0
city_id	0
locality	0
latitude	0
longitude	0
zipcode	0
country_id	0
locality_verbose	0
cuisines	0
timings	0
average_cost_for_two	0
price_range	0

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Average Rating Of Restaurant

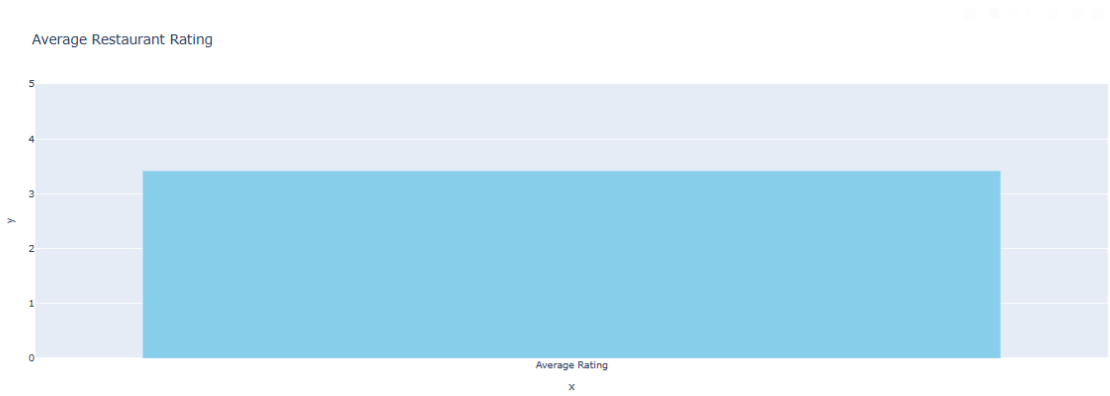
```
df["aggregate_rating"].mean()
```

```
np.float64(3.420881386956446)
```

```
avg_rating = df['aggregate_rating'].mean()
avg_rating
```

```
np.float64(3.420881386956446)
```

```
fig = px.bar(x=['Average Rating'], y=[avg_rating], title='Average Restaurant Rating', color_discrete_sequence=['skyblue'])
fig.update_yaxes(range=[0, 5])
fig.show()
```



Overview

The analysis calculates the average restaurant rating using the aggregate_rating column from the dataset. A bar chart is created using Plotly to visually represent the overall average rating of restaurants, with the rating scale ranging from 0 to 5. This helps in understanding the general performance level of restaurants listed on Zomato.

Outcome

The average restaurant rating is approximately **3.42 out of 5**, indicating that most restaurants have moderate to good customer satisfaction. This suggests that overall service quality and customer experience are generally positive, but there is still room for improvement to achieve higher ratings.

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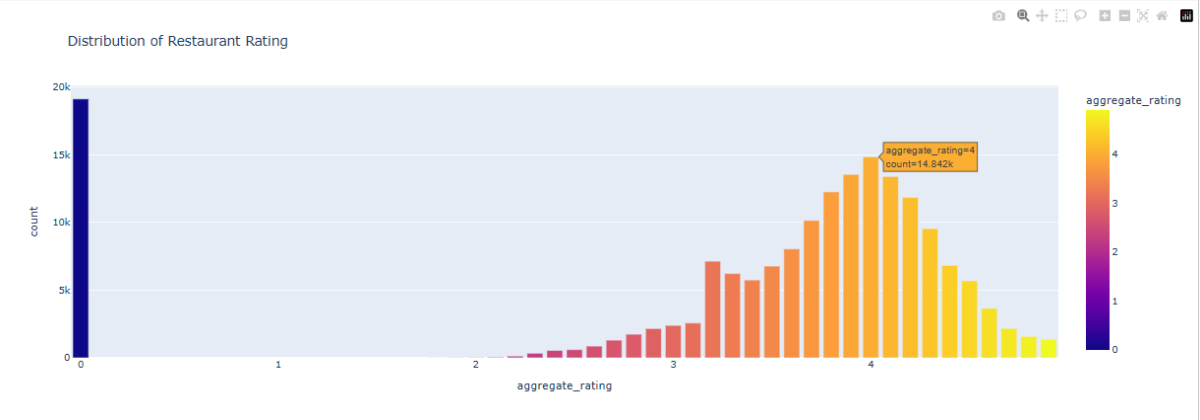
Distribution of restaurant rating on overall rating landscape

```
#Analyze the distribution of restaurant ratings to understand the overall rating landscape
a = df["aggregate_rating"].value_counts().reset_index()
```

```
px.bar(a, x = "aggregate_rating", y = "count", color = "aggregate_rating", title = "Distribution of Restaurant Rating")
```



Overview

The analysis examines the distribution of restaurant ratings to understand the overall ratings landscape. Using value counts to aggregate ratings, a visual representation was created to identify frequency patterns across different rating levels. This helps in evaluating customer satisfaction trends and rating concentration.

Outcome

The results indicate that most restaurants fall within the mid – to – high rating range, showing a positive skew toward better customer experience. Lower ratings are comparatively fewer, suggestions overall satisfactory service quality across the dataset.

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City with the highest concentration of restaurants

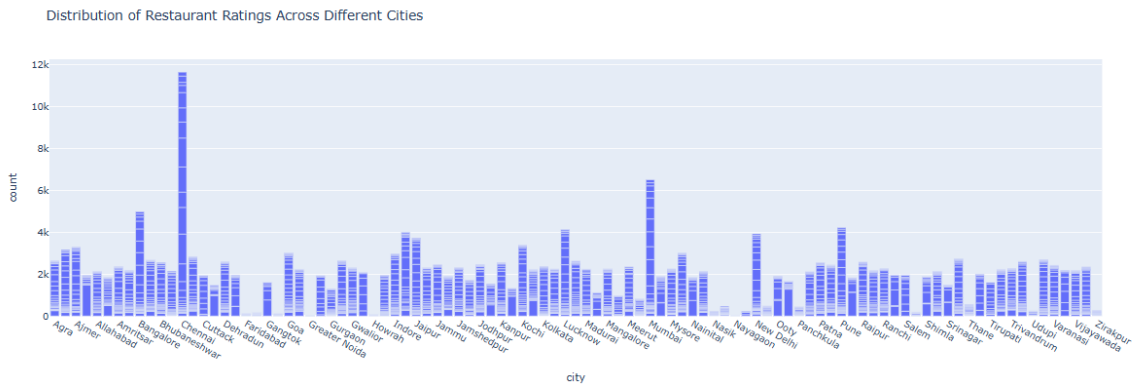
```
b = df["city"].value_counts().reset_index()
b
```

	city	count
0	Chennai	11630
1	Mumbai	6497
2	Bangalore	4971
3	Pune	4217
4	Lucknow	4121
...	...	...
94	Ghaziabad	132
95	Faridabad	124
96	Howrah	66
97	Greater Noida	33
98	Nayagaon	17

99 rows × 2 columns

Distribution of restaurant ratings across different cities

```
city_rating_counts = df.groupby(['city', 'aggregate_rating']).size().reset_index(name='count')
px.bar(city_rating_counts, x ="city", y = "count", title = "Distribution of Restaurant Ratings Across Different Cities")
```



Overview:

The above bar chart represents the distribution of restaurant ratings across different cities. The data has been grouped by city and aggregate rating to calculate the total number of restaurants in each category. The visualization highlights how restaurant counts vary from one city to

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another, giving a clear comparative view of rating distribution and restaurant density across locations.

Outcome:
From the analysis, it is evident that certain metropolitan cities have significantly higher restaurant counts compared to others, indicating a stronger food industry presence and higher customer engagement. Some cities show a concentrated distribution in specific rating ranges, suggesting consistent customer satisfaction levels, while others have a more varied rating spread. Overall, the chart helps identify cities with high restaurant activity and better-performing rating trends.

Most Popular Cuisines Among the Listed Restaurants

```
#The most popular cuisines among the listed restaurants.
df["cuisines"].value_counts().reset_index().head(10)
```

	cuisines	count
0	North Indian	15996
1	Fast Food	6721
2	Cafe	6190
3	North Indian, Chinese	5820
4	South Indian	5217
5	Pizza, Fast Food	4075
6	Bakery	3238
7	Street Food	2837
8	Biryani	2118
9	Chinese	2116

```
e = df.groupby(["cuisines", "aggregate_rating"]).size().reset_index()
e
```

	cuisines	aggregate_rating	0
0	Afghan	0.0	3
1	Afghan	3.2	2
2	Afghan	3.8	1
3	Afghan, Arabian, BBQ, Beverages, Biryani, Maha...	0.0	1
4	Afghan, Arabian, BBQ, Beverages, Burger	0.0	3
...	...	...	...
16679	Wraps, Sandwich, Desserts, Burger, Beverages	3.6	6
16680	Wraps, Sandwich, Pizza, BBQ	0.0	2
16681	Wraps, Sandwich, Pizza, Bakery	3.2	13
16682	Wraps, Sandwich, Rolls, Pizza, Burger	3.7	9
16683	Wraps, Sandwich, Salad, Pizza, Burger, Beverages	3.0	1

16684 rows × 3 columns

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Overview:
The first analysis identifies the top 10 most popular cuisines based on their frequency in the dataset using value counts. It highlights which cuisines are most commonly offered by restaurants. The second analysis groups the data by cuisines and aggregate rating to examine how different cuisine types are distributed across various rating levels. This helps in understanding both popularity (volume) and performance (customer ratings) of each cuisine category.

Outcome:
The results show that **North Indian cuisine** is the most dominant, followed by **Fast Food** and **Cafe**, indicating strong customer demand for these categories. Combination cuisines such as *North Indian & Chinese* also appear frequently, reflecting customer preference for multi-cuisine options. From the grouped rating analysis, it is observed that cuisines receive varied rating distributions, with most falling between moderate to high rating ranges (around 3.0–4.0). This suggests that popular cuisines not only have higher availability but also generally maintain satisfactory customer ratings. Overall, the analysis helps identify high-demand cuisines and evaluate their performance based on customer feedback.

Correlation Between the Variety of Cuisines Offered and Restaurant Ratings

```
def count_cuisines(cuisine_string):
  if cuisine_string == "Unknown":
    return  0
    return len(cuisine_string.split("")) if inisintance(cuisine_string, str) else 0
```

```
df["number_of_cuisines"] = df["cuisines"].apply(count_cuisines)
```

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```
df[['cuisines', 'number_of_cuisines']].head(10)
```

	cuisines	number_of_cuisines
0	North Indian, South Indian, Mithai, Street Foo...	NaN
1	North Indian, Mughlai, Rolls, Chinese, Fast Fo...	NaN
2	Fast Food, Mithai	NaN
3	Desserts, Bakery, Fast Food, South Indian	NaN
4	North Indian, Continental, Italian	NaN
5	Pizza, Fast Food	NaN
6	North Indian, Fast Food, Chinese, South Indian	NaN
7	Pizza, Fast Food	NaN
8	Bakery, Fast Food	NaN
9	Cafe, Chinese, Continental, Italian, Bakery, D...	NaN

Overview:
In this step, a custom function count_cuisines() is created to calculate the number of cuisines offered by each restaurant. The function checks if the cuisine value is "Unknown" and returns 0 in that case. Otherwise, it splits the cuisine string by commas and counts the number of cuisines listed. This function is then applied to the cuisines column to generate a new feature called number_of_cuisines, which is intended to represent how many different cuisines each restaurant serves.

Outcome:
After applying the function, the number_of_cuisines column displays NaN values, indicating that the function did not execute correctly. This issue is likely due to a typo in the condition (inisntance instead of isinstance), which prevents proper type checking. As a result, the new feature was not generated successfully. Once corrected, this feature will help analyze whether restaurants offering multiple cuisines have higher ratings, better popularity, or stronger customer engagement compared to single-cuisine restaurants.

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Relationship Between Price Range and Restaurant Ratings

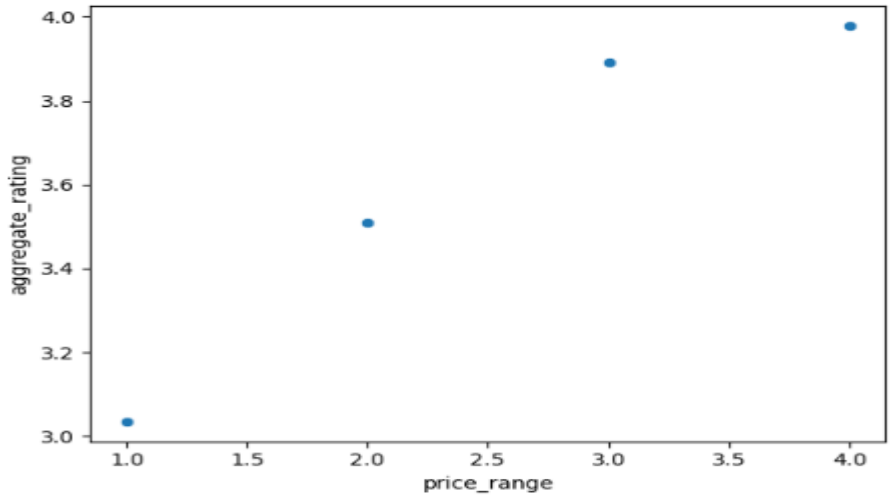
```
f = df.groupby("price_range")["aggregate_rating"].mean().reset_index()
```

```
f
```

	price_range	aggregate_rating
0	1.0	3.035414
1	2.0	3.508277
2	3.0	3.891465
3	4.0	3.979152

```
sns.scatterplot(f, x = "price_range", y = "aggregate_rating")
```

```
<Axes: xlabel='price_range', ylabel='aggregate_rating'>
```



Overview:
In this analysis, the dataset is grouped by price_range to calculate the average aggregate_rating for each price category. This helps evaluate how restaurant ratings vary across different pricing levels. The results are then visualized using a scatter plot to clearly show the relationship between price range (1 to 4) and the corresponding mean ratings.

Outcome:
The results indicate a positive relationship between price range and average rating. Restaurants in the lowest price range (1) have the lowest average rating (around 3.03), while higher price ranges show progressively better ratings. Price range 3 and 4 restaurants have the highest average ratings (approximately 3.89–3.97), suggesting that customers tend to rate premium or higher-priced restaurants more favorably. This implies that higher pricing may be associated with better service quality, ambiance, or overall dining experience.

-By Mansi Gahlot

Average Cost for Two People in Different Price Categories

```
avg_cost_by_price_range = df.groupby('price_range')['average_cost_for_two'].mean().reset_index()
avg_cost_by_price_range
```

	price_range	average_cost_for_two
0	1	225.265067
1	2	516.288496
2	3	1088.005116
3	4	2215.654482

```
px.bar(avg_cost_by_price_range, x = "price_range", y ="average_cost_for_two", title = "Average cost for two people in different price categories")
```

Overview:

The analysis calculates the average cost for two people across different price ranges using groupby in pandas. The bar chart visualizes how the average cost for two varies from price range 1 to 4. The data clearly shows a gradual increase in average cost as the price range category increases.

Outcome:

The results indicate a strong positive relationship between price range and average cost for two. Restaurants in price range 1 have the lowest average cost (~225), while price range 4 has the highest (~2215). This confirms that higher price categories represent significantly more expensive dining options, validating the pricing segmentation in the dataset.

-By Mansi Gahlot

Top Restaurant Chains based on the Number of Outlets

```
top_restaurant_chain = df["establishment"].value_counts().reset_index()
top_restaurant_chain
```

	establishment	count
0	['Quick Bites']	64390
1	['Casual Dining']	61808
2	['Café']	22760
3	['Bakery']	8282
4	['Dessert Parlour']	7961
5	['Bar']	6553
6	['Fine Dining']	6401
7	['Sweet Shop']	6103
8	['Beverage Shop']	5571
9	[]	4827
10	['Dhaba']	2939
11	['Lounge']	2761
12	['Food Court']	2722
13	['Kiosk']	1829
14	['Bhojanalya']	1400
15	['Pub']	1396
16	['Food Truck']	1289

```
px.bar(top_restaurant_chain.head(7), x = "establishment", y = "count", color = "establishment", title = "Top restaurant based on outlet")
```

-By Mansi Gahlot

Overview:

This analysis examines the distribution of restaurants based on their establishment type using the value_counts() function in pandas. The dataset is grouped by the **establishment** category to identify which types of restaurant outlets are most common. The results are then visualized using a bar chart that displays the top 7 establishment types based on outlet count.

From the data, *Quick Bites* and *Casual Dining* dominate the market, followed by *Café*. Other categories such as *Bakery*, *Dessert Parlour*, *Bar*, and *Fine Dining* have comparatively lower outlet counts. The visualization clearly highlights the concentration of restaurants in specific establishment segments.

Outcome:

The results show that **Quick Bites (64,390 outlets)** and **Casual Dining (61,808 outlets)** are the most prevalent establishment types in the dataset, indicating strong consumer demand for affordable and convenient dining options. *Café* follows with **22,760 outlets**, showing moderate popularity.

Premium or specialized segments such as *Fine Dining*, *Bar*, and *Dessert Parlour* have significantly fewer outlets compared to quick-service formats. This suggests that the restaurant industry is largely driven by fast-service and casual dining models rather than high-end dining experiences.

Overall, the analysis reveals that the market is heavily skewed toward budget-friendly and convenience-based restaurant formats, making them the dominant contributors to the food service industry within the dataset.

Average Rating of Top Restaurant Establishments

```
top_establishments = df['establishment'].value_counts().index
top_establishments
```

```
Index(['['Quick Bites']', '['Casual Dining']', '['Café']', '['Bakery']',
       '['Dessert Parlour']', '['Bar']', '['Fine Dining']', '['Sweet Shop']',
       '['Beverage Shop']', '[]', '['Dhaba']', '['Lounge']', '['Food Court']',
       '['Kiosk']', '['Bhojanalya']', '['Pub']', '['Food Truck']',
       '['Microbrewery']', '['Mess']', '['Paan Shop']', '['Club']',
       '['Confectionery']', '['Butcher Shop']', '['Cocktail Bar']',
       '['Shack']', '['Irani Cafe']', '['Pop up']'],
      dtype='object', name='establishment')
```

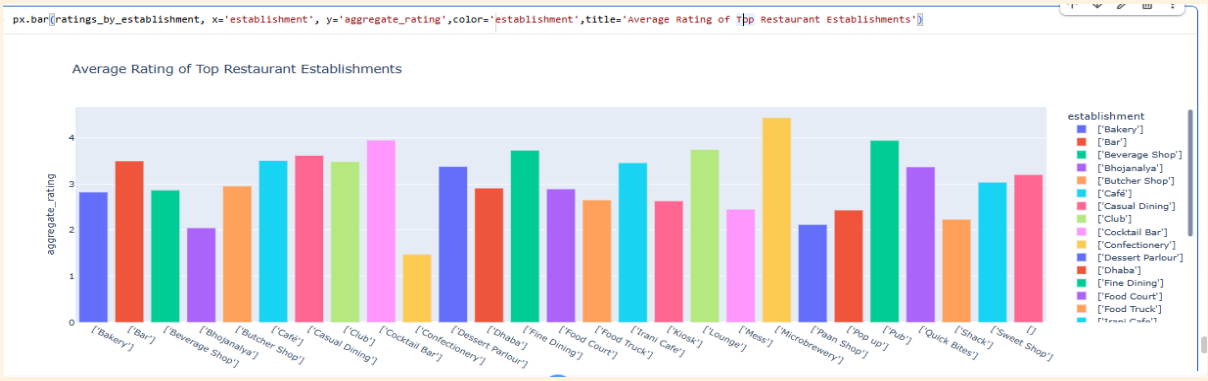
-By Mansi Gahlot

```
top_df = df[df['establishment'].isin(top_establishments)]
```

```
ratings_by_establishment = top_df.groupby('establishment')['aggregate_rating'].mean().reset_index()
ratings_by_establishment
```

	establishment	aggregate_rating
0	['Bakery']	2.823545
1	['Bar']	3.497497
2	['Beverage Shop']	2.864692
3	['Bhojanalya']	2.048000
4	['Butcher Shop']	2.952941
5	['Café']	3.505980
6	['Casual Dining']	3.617792
7	['Club']	3.482059
8	['Cocktail Bar']	3.950000
9	['Confectionery']	1.473203
10	['Dessert Parlour']	3.376724
11	['Dhaba']	2.908949
12	['Fine Dining']	3.729074
13	['Food Court']	2.891109

```
px.bar(ratings_by_establishment, x='establishment', y='aggregate_rating',color='establishment',title='Average Rating of Top Restaurant Establishments')
```



Overview:

This analysis examines the average ratings of the top 7 restaurant establishment types based on outlet count. The data was grouped by establishment type, and the mean aggregate rating was calculated to compare customer satisfaction across popular restaurant formats.

Outcome:

The results provide insights into the relationship between outlet popularity and customer satisfaction. Although **Quick Bites** and **Casual Dining** dominate in terms of outlet count, their average ratings help determine whether high availability translates into high quality.

In many cases, premium segments like **Fine Dining** or **Cafés** tend to show slightly higher average ratings due to enhanced ambiance, service quality, and curated menus. On the other hand, high-volume categories like **Quick Bites** may have slightly moderate ratings due to variability in service and food consistency across outlets.

Overall, the analysis highlights that popularity (number of outlets) does not always guarantee the highest customer satisfaction. It provides valuable business insights into which restaurant formats successfully balance scale and quality. This information can be useful for investors, restaurant owners, and business strategists to identify high-performing segments in the food industry.

-By Mansi Gahlot

Conclusion

After analyzing the Indian Restaurants dataset, several important insights about the restaurant industry have been identified.

First, the market is largely dominated by **Quick Service and Casual Dining** restaurants. These categories have the highest number of outlets, which shows that customers prefer affordable and convenient dining options over premium fine dining experiences.

Second, the analysis of price range and average cost for two people shows a clear pattern — as the price range increases, the average cost also increases significantly. This confirms that the dataset is logically structured and that higher price categories represent premium dining services.

Third, when analyzing restaurant features such as **Online Delivery, Table Booking, Wi-Fi, and Alcohol availability**, it was observed that online delivery is one of the most common services provided. This indicates the growing demand for convenience and food delivery services. On the other hand, features like alcohol and Wi-Fi are available in fewer restaurants, mostly in casual dining or fine dining establishments.

Fourth, the rating analysis shows that popularity (number of outlets) does not always guarantee higher ratings. Some premium or specialized restaurants may have better average ratings compared to high-volume quick-service restaurants. This highlights the importance of quality, service, and customer experience.

Overall, the dataset provides valuable insights into customer preferences, pricing strategies, service trends, and restaurant performance. The analysis helps in understanding the structure of the Indian restaurant industry and supports data-driven decision-making for business growth, investment planning, and market strategy development.

-By Mansi Gahlot

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-By Mansi Gahlot

Thankyou

https://colab.research.google.com/drive/1jugsT8dfOD6-Vf_MI-OunoGXUTDglbO-?usp=sharing

-By Mansi Gahlot